Vance Vanvolkenburgh
655.662 — Introduction to Healthcare Systems Engineering
Module 7 | M7A2: Software Requirements

Software Requirements — Peritoneal Dialysis Remote Patient Monitoring (PD-RPM)

On the course taxonomy PD-RPM is a **software-intensive** system: sensors and communications hardware with more than 40 percent of its capability in software, modeled in SysML and UML. The five requirements below are therefore allocated to named software elements of the M5A1 physical architecture rather than to the system as a whole, and each is written at the level where software requirements are properly identified — detailed design, levels 5 and 6 — not at the top level, which stays implementation-free. Each states a single **shall**, is testable by a named method, and specifies software functionality, a constraint, or a performance criterion.

Req	Software element (M5A1)	Requirement class	What it governs
SR-1	Analytics Engine (AE-1)	Functionality	Ultrafiltration trend computation
SR-2	Rules and Alert Engine (RE-1)	Performance	Alert evaluation and suppression
SR-3	Ingest and Normalization (IS-1)	Functionality	Payload validation and fault containment
SR-4	Identity and Access Management (IM-1)	Constraint	Access control and audit
SR-5	All PD-RPM software items	Constraint	Software lifecycle and release gate

SR-1 — Ultrafiltration trend computation · Analytics Engine (AE-1) · Functionality requirement

The Analytics Engine (AE-1) shall compute per-session net ultrafiltration for each of the trailing 14 validated treatment sessions and shall set a declining-trend flag when the 7-day mean falls more than 20 percent below the trailing 30-day mean, completing the computation within 10 minutes of a session being persisted to the Data Store (DS-1), suppressing the flag when fewer than 8 of the trailing 14 sessions are present, and producing identical output for identical input.

Allocated from: OR-1, derived from operational need ON-1
Verification: Test by replay of 60 recorded patient-months against a reference implementation, including sparse, out-of-order, and duplicate-session cases

SR-2 — Alert evaluation and suppression · Rules and Alert Engine (RE-1) · Performance requirement

The Rules and Alert Engine (RE-1) shall evaluate the complete active rule set against each newly persisted record within 60 seconds of persistence at a sustained load of 150 records per minute, shall dispatch each resulting critical alert to the Notification Gateway (NG-1) within a further 60 seconds, and shall emit no more than one alert per patient per rule in any rolling 24-hour period.

Allocated from: OR-3; KPP Alert Latency (15 minutes end to end, of which this allocates 2); KPP False Alert Rate (≤ 1 non-actionable alert per patient per month)
Verification: Test at 150 records per minute for 4 hours with timestamps captured at RE-1 and NG-1; suppression verified by injecting repeated trigger conditions

SR-3 — Payload validation and fault containment · Ingest and Normalization (IS-1) · Functionality requirement

The Ingest and Normalization service (IS-1) shall validate every inbound payload against the schema version declared in that payload's header, shall convert all masses to kilograms and all volumes to milliliters before persistence, shall route any payload that fails validation or carries a value outside its configured physiological range to a dead-letter queue within 2 seconds of receipt, and shall not discard a payload or substitute a default value for any rejected field.

Allocated from: OR-2; KPP Data Completeness (95 percent of sessions ingested within 24 hours)
Verification: Test by schema fuzzing and boundary-value injection at each range limit, with dead-letter queue counts reconciled against the injected fault set

SR-4 — Access control and audit · Identity and Access Management (IM-1) · Constraint requirement

The Identity and Access Management service (IM-1) shall authenticate every clinician session using multi-factor authentication, shall restrict each authenticated principal to the patient panel assigned to that principal's role, shall terminate any session idle for more than 15 minutes, and shall write to append-only storage an audit record of every read or write of patient data containing the principal identifier, patient identifier, UTC timestamp, and action performed, retained for not less than 6 years.

Allocated from: OR-4; HIPAA Security Rule access control and audit control standards (45 CFR §164.312)
Verification: Test by penetration testing of role boundaries and session expiry, plus reconciliation of the audit log against a scripted access sequence of known length

SR-5 — Software lifecycle and release gate · All PD-RPM software items · Constraint requirement

All PD-RPM software items shall be developed, maintained, and documented in accordance with IEC 62304 software safety Class B, and each release candidate shall carry a machine-readable software bill of materials in SPDX 2.3 or CycloneDX 1.5 format enumerating every third-party component and its version, and shall carry no unresolved static or dynamic analysis finding scored CVSS 7.0 or higher.

Allocated from: IEC 62304; ANSI/AAMI SW96 and the FDA 2023 premarket cybersecurity guidance, which together convert the software lifecycle process into a Secure Product Development Framework
Verification: Inspection of the IEC 62304 documentation set at CDR, and demonstration of the automated SBOM generation and SAST/DAST gate failing a build seeded with a known high-severity component

On the Class B claim in SR-5. IEC 62304 assigns a software safety class by the injury that software failure can contribute to. PD-RPM does not command therapy: the cycler sits outside the system boundary and the nephrologist holds clinical decision authority. A failure here therefore cannot directly injure a patient, but a missed or delayed peritonitis alert can contribute to non-serious or serious injury through delayed treatment, which places the software above Class A and below Class C. Stating the class and the reasoning for it is what makes SR-5 auditable rather than aspirational.

Two boundaries are deliberate. These requirements say nothing about the interfaces between these elements, which M7A1 covers separately, because the course treats interface definition as its own layer with its own requirements. And none of them states a top-level system behavior: SR-1 through SR-4 are each traceable upward to an operational requirement or KPP from M3A1 that is itself implementation-free, which is the flow-down the module describes. The patient tablet application remains bounded only by OR-3 and the 60-second interaction KPP and has not yet had its software requirements written; that is the next tranche, not an omission from this set.